Enumeration of Polygon-Free Mosaics

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Abstract

Hong and Oh introduced a model for multiple ring polymers in physics in 7 distinct tiles are used to form an $m \times n$ retangular grid, which they call *mosaics*. Hong and Oh provided bounds on the number of these mosaics in which the tile patterns form polygons and all tiles that are not part of a polygon are the blank tile, which they call *polygon mosaics*. We exactly enumerate polygon mosaics. We also introduce and enumerate mosaics with the related property of not containing a polygon, which we call polygon-free mosaics.

1 Mosaics

Imagine you are tasked with tiling a rectangular bathroom floor that is m units by n units, blindfolded. At your disposal is an unending supply of 7 distinct types of tiles. These tiles, diagrammed in Figure 1, are composed of unit squares with red dotted lines connecting 2 sides at their midpoints, as well as the "blank" tile T_0.

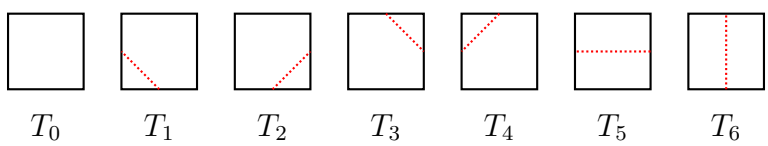


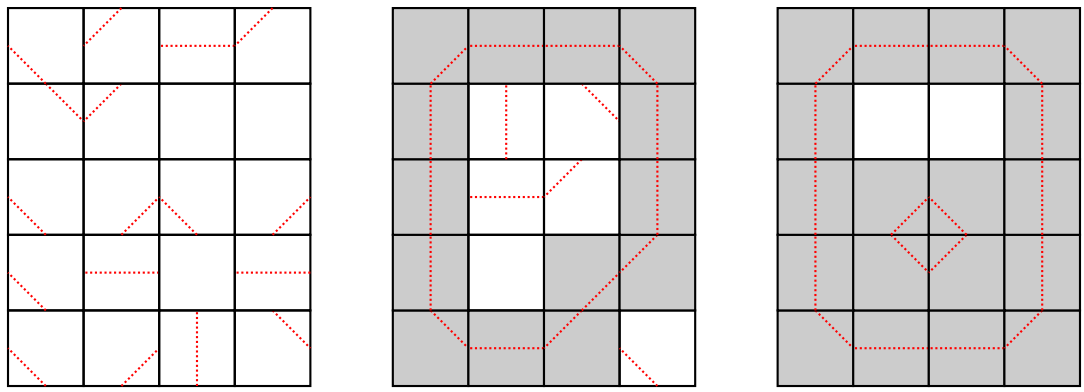
Figure 1: The tile set $\mathbb{T}$

We denote the set of tiles $\mathbb{T} = \{T_0, \ldots, T_6\}$. The task is complete once you place mn tiles from $\mathbb{T}$ to cover the floor, after which you remove your blindfold. We are interested in the number of floors that do not contain polygons[1].

We call a fully tiled $m \times n$ floor an (m, n) *mosaic*, where m, n are positive integers. Figure 2 shows various $(5, 4)$ mosaics. We highlight the tiles that are part of a polygon in gray.

Definition 1.1. An (m, n) *polygon-free mosaic* is an (m, n) mosaic that does not contain a polygon. An (m, n) *polygon mosaic* is an (m, n) mosaic such that any tile that is not part of a polygon is T_0.

[1]Polygons are more commonly referred to as "self-avoiding polygons" in the literature[2, 3] to highlight their connection with self-avoiding walks.



(a) A polygon-free mosaic (b) A mosaic that is not polygon-free (c) A polygon mosaic

Figure 2: Examples of $(5, 4)$ mosaics

Figure 2a depicts a polygon-free mosaic, Figure 2b depicts a mosaic that is not polygon-free, and Figure 2c depicts a polygon mosaic. Notice that the only mosaic that is both a polygon mosaic and a polygon-free mosaic is the mosaic of all T_0 tiles.

Let $\mathbb{P}^{(m,n)}$ be the set of (m, n) polygon-free mosaics. Clearly the number of (m, n) mosaics that contain at least one polygon is then $7^{mn} - \left|\mathbb{P}^{(m,n)}\right|$. To determine the number of polygon-free mosaics, we begin by defining a related object to mosaics, that being binary lattices.

2 Binary Lattices

For positive integers m, n, an (m, n) *binary lattice* is a rectangular lattice of $m + 1$ by $n + 1$ vertices, with each vertex labeled 0 or 1. We also define a *framed lattice* to be a binary lattice in which the boundary vertices are labeled 0. An example of a $(5, 7)$ framed lattice is shown on the right of Figure 3.

Definition 2.1. Let a *cell* be a $(1, 1)$ binary lattice.

When referring to cells, we omit the connecting lines. There are sixteen cells:

$$\begin{smallmatrix}0&0\\0&0\end{smallmatrix},\ \begin{smallmatrix}0&0\\0&1\end{smallmatrix},\ \begin{smallmatrix}0&0\\1&0\end{smallmatrix},\ \begin{smallmatrix}0&0\\1&1\end{smallmatrix},\ \begin{smallmatrix}0&1\\0&0\end{smallmatrix},\ \begin{smallmatrix}0&1\\0&1\end{smallmatrix},\ \begin{smallmatrix}0&1\\1&0\end{smallmatrix},\ \begin{smallmatrix}0&1\\1&1\end{smallmatrix},\ \begin{smallmatrix}1&0\\0&0\end{smallmatrix},\ \begin{smallmatrix}1&0\\0&1\end{smallmatrix},\ \begin{smallmatrix}1&0\\1&0\end{smallmatrix},\ \begin{smallmatrix}1&0\\1&1\end{smallmatrix},\ \begin{smallmatrix}1&1\\0&0\end{smallmatrix},\ \begin{smallmatrix}1&1\\0&1\end{smallmatrix},\ \begin{smallmatrix}1&1\\1&0\end{smallmatrix},\ \text{and}\ \begin{smallmatrix}1&1\\1&1\end{smallmatrix}.$$

Definition 2.2. Let a *proper lattice* be a binary lattice that does not contain any $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ cells.

Definition 2.3. Let $\mathcal{L}$ map a (m, n) polygon mosaic to a (m, n) proper framed lattice with the following rule. If the vertex is surrounded by the dotted lines of an even number of polygons (including 0 polygons), label it 0. If the vertex is surrounded by the dotted lines of an odd number of polygons, label it 1. Removing the dotted lines from the tiles gives the framed lattice.

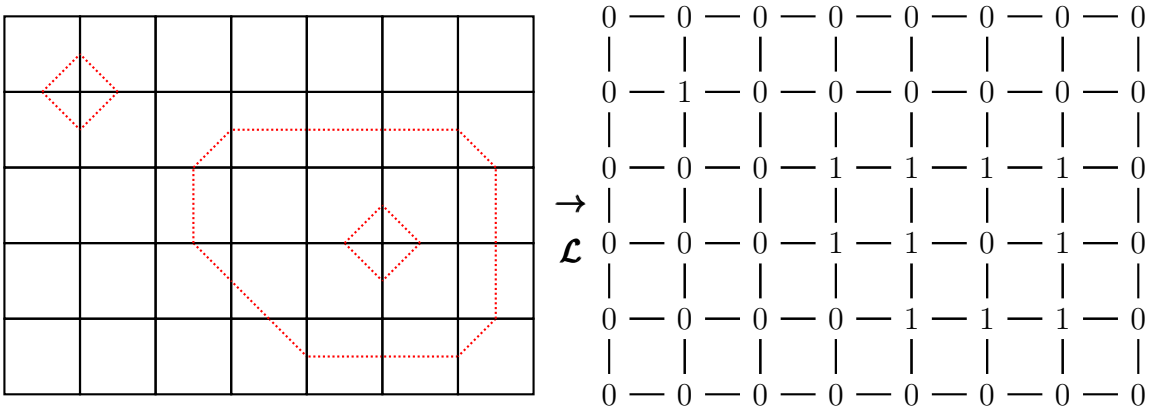


Figure 3: $\mathcal{L}$ applied to a polygon mosaic

Notice that under $\mathcal{L}$, no tile can map to the cells $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$.

Proposition 2.1. *The map $\mathcal{L}$ bijectively maps the set of (m, n) polygon mosaics to the set of (m, n) proper framed lattices for all positive integers m and n.*

Proof. The map $\mathcal{L}$ is injective no two polygon mosaics can map to the same binary lattice. Simply pick a tile for which the two polygon mosaics differ. From the definition of $\mathcal{L}$, the corresponding cell in the framed binary lattice must be different.

The map $\mathcal{L}$ is surjective because each framed proper lattice is mapped to by one polygon mosaic. This polygon mosaic can be constructed by replacing each cell of the lattice with the unique tile in $\mathbb{T}$ that maps to that cell under $\mathcal{L}$. The mosaic constructed from this procedure is a polygon mosaic because **TODO**

□

3 Counting Polygon Mosaics

To enumerate polygon mosaics, we begin by defining a function v that maps cells to integers. More generally, we define v from binary lattices to integers by taking the product of v applied to each cell in the binary lattice.

We say we *assign* values to v if we substitute specific integers for v of each cell. We choose this construction because we will show that just by changing the assignment of integers in v, we can enumerate both polygon mosaics (Section 3) and polygon-free mosaics (Section 5) under the tile set $\mathbb{T}$, as well as certain extensions to $\mathbb{T}$ (Section 6).

Remark 3.1. If we assign

$$\begin{cases} 0 & \text{for cells } \begin{smallmatrix}0&1\\1&0\end{smallmatrix}, \begin{smallmatrix}1&0\\0&1\end{smallmatrix} \\ 1 & \text{otherwise} \end{cases}$$

The number of (m, n) polygon mosaics is $\sum v(\ell)$, where the sum is over all framed lattices ℓ.

Proof. From Proposition 2.1, there is a bijection between polygon mosaics and framed proper lattices, and so $v(\ell)$ counts one poylgon mosaic for each framed proper lattice ℓ. By the definition of v, if ℓ' is a binary lattice that contains either $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ or $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, then $v(\ell') = 0$. Therefore, $\sum v(\ell)$ over all framed lattices is equal to the sum over all proper framed lattices. $\square$

We show that we can more efficiently compute the sum $\sum v(\ell)$ from Remark 3.1 using matrices.

4 Computing $\sum v(\ell)$ with Matrices

In this section we provide no assignment for v, and so represent assignments of cells just as v of the cell. For example, $v\left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ represents the arbitrary assignment for cell $\left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$.

Define $A_1 = \left(v\left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)\right), B_1 = \left(v\left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)\right), C_1 = \left(v\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)\right), D_1 = \left(v\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)\right)$, and for positive integer n,

$$A_{n+1} = \begin{pmatrix} v\left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)A_n & v\left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)B_n \\ v\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)C_n & v\left(\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}\right)D_n \end{pmatrix} \quad B_{n+1} = \begin{pmatrix} v\left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)A_n & v\left(\begin{smallmatrix} 0 & 0 \\ 1 & 1 \end{smallmatrix}\right)B_n \\ v\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)C_n & v\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)D_n \end{pmatrix}$$

$$C_{n+1} = \begin{pmatrix} v\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)A_n & v\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)B_n \\ v\left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)C_n & v\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)D_n \end{pmatrix} \quad D_{n+1} = \begin{pmatrix} v\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)A_n & v\left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)B_n \\ v\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)C_n & v\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)D_n \end{pmatrix}.$$

Definition 4.1. Let the n digit binary representation of the nonnegative integer $k < 2^n$ be written as $\beta_n(k)$, with $\beta_0(0)$ being the empty string.

We extend our shorthand for cells to $(1, n)$ binary lattice using analagous matrix notation and the definition for $\beta_n(k)$. For example, we write

$$v\begin{pmatrix} 0 & \beta_2(1) & 0 \\ 0 & \beta_2(0) & 0 \end{pmatrix} = v\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = v\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} v\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

For matrices, mosaics, and lattices, we number the rows $0, 1, \ldots, m-1$ top to bottom, and columns $0, 1, \ldots, n-1$ left to right.

Proposition 4.1. *For positive integer n and nonnegative integers $i, j < 2^{n-1}$, the (i, j)-th entry of*

$$A_n \text{ is } v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix}, \quad B_n \text{ is } v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 1 & \beta_{n-1}(j) & 0 \end{pmatrix},$$
$$C_n \text{ is } v\begin{pmatrix} 1 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix}, \quad D_n \text{ is } v\begin{pmatrix} 1 & \beta_{n-1}(i) & 0 \\ 1 & \beta_{n-1}(j) & 0 \end{pmatrix}.$$

Proof. We prove by induction on n. The $n = 1$ case is trivial, as $\beta_0(0)$ is the empty string. We next assume our result for some fixed $n \geq 1$. The full induction step requires proving the entires in matrices $A_{n+1}, B_{n+1}, C_{n+1}$, and D_{n+1} all follow the proposition for each of the four $2^{n-1} \times 2^{n-1}$ block matrices that define them. We do two of the sixteen cases for illustration, that being the upper right block matrix in A_{n+1} then the lower left block matrix in D_{n+1}.

Let i, j be nonnegative integers less than 2^{n-1} that index entries in A_n, B_n, C_n, or D_n. For the upper right block matrix in A_{n+1}, by definition the $(i, 2^{n-1} + j)$ entry of A_{n+1} equals $v\left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ times the (i, j) entry of B_n. By our induction hypothesis, this equals

$$v\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 1 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 0 & 0 & \beta_{n-1}(i) & 0 \\ 0 & 1 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 0 & \beta_n(i) & 0 \\ 0 & \beta_n(2^{n-1}+j) & 0 \end{pmatrix},$$

as desired.

Similarly, for the lower left block matrix in D_{n+1}, by definition the $(2^{n-1}+i, j)$ entry of D_{n+1} equals $v\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ times the (i, j) entry of C_n. By our induction hypothesis, this equals

$$v\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} v\begin{pmatrix} 1 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 1 & 1 & \beta_{n-1}(i) & 0 \\ 1 & 0 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 1 & \beta_n(2^{n-1}+i) & 0 \\ 1 & \beta_n(j) & 0 \end{pmatrix},$$

again as desired. These arguments must be adapted to prove the desired results for all blocks of $A_{n+1}, B_{n+1}, C_{n+1}$, and D_{n+1}, but all the work is similar. □

Theorem 4.2. *For all positive integers m and n, the $(0,0)$ entry of A_n^m equals $\sum v(\ell)$, where the sum is over all (m,n) framed lattices ℓ.*

Proof. Fix n. We first introduce some terminology. For all positive integers m, we say a *height m strip* is an (m,n) binary lattice in which all vertices in the outside columns (i.e., column 0 and column n) are labeled zero. Then for a height m strip ℓ and a nonnegative integer $i < 2^{n-1}$, we say a particular row of ℓ is *determined by i* if that row is equal to $0\ \beta_{n-1}(i)\ 0$. Note that an (m,n) framed lattice is exactly a height m strip in which both the top and bottom rows are determined by 0. Thus, to prove our proposition it suffices to show the following.

Claim. For all positive integers m and nonnegative integers $i, j < 2^{n-1}$, the (i,j) entry of A_n^m equals $\sum v(\ell)$, where the sum is over all height m strips ℓ whose first row is determined by i and whose last row is determined by j.

To prove the claim, first observe that for each i and j, there exists a unique height 1 strip whose first row is determined by i and whose last row is determined by j. By Proposition 4.1, v applied to this strip is equal to the (i,j) entry of A_n. This proves our claim when $m = 1$. We continue by induction, so for some $m \geq 2$, assume the claim for $m-1$.

Fix nonnegative integers $i, j < 2^{n-1}$, and consider some nonnegative integer $k < 2^{n-1}$. As in the $m = 1$ case, Proposition 4.1 tells us that the (i,k) entry of A_n equals v applied to the unique height 1 strip with first and last rows determined respectively by i and by k. By our induction hypothesis, the (k,j) entry of A_n^{m-1} equals $\sum v(\ell)$, where the sum is over all height $m-1$ strips ℓ whose first and last rows are determined respectively by k and by j. Consider applying v both to the height 1 strip indicated above and to one of the height $m-1$ strips ℓ in this sum. As in Equation 1, by the definition of v, the product of these two numbers is equal to v applied to the height m strip formed by overlapping the bottom row of the height 1 strip to the top row of ℓ.

$$v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(k) & 0 \end{pmatrix} v\begin{pmatrix} 0 & \beta_{n-1}(k) & 0 \\ \vdots & \vdots & \vdots \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(k) & 0 \\ \vdots & \vdots & \vdots \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix} \tag{1}$$

From this we see that the product of the (i,k) entry of A_n and the (k,j) entry of A_n^{m-1} is equal to $\sum v(\ell)$, where the sum is now over all height m strips ℓ whose first, second, and last rows are determined respectively by i, by k, and by j. This tells us, in turn, that when we additionally sum over all nonnegative $k < 2^{n-1}$, we can conclude that the (i,j) entry of

$A_n \cdot A_n^{m-1} = A_n^m$ is, as desired, the sum $\sum v(\ell)$ where the sum is over all height m strips ℓ whose first and last rows are determined respectively by i and by j. $\square$

Assigning the values for v from Remark 3.1 into the matrices from Theorem 4.2 gives the following theorem.

Theorem 4.3. *For positive integers n let A_n, B_n, C_n, D_n be $2^{n-1} \times 2^{n-1}$ matrices with integer entries, where $A_1 = B_1 = C_1 = D_1 = \begin{pmatrix} 1 \end{pmatrix}$ and*

$$A_{n+1} = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix} \quad B_{n+1} = \begin{pmatrix} A_n & B_n \\ 0C_n & D_n \end{pmatrix}$$
$$C_{n+1} = \begin{pmatrix} A_n & 0B_n \\ C_n & D_n \end{pmatrix} \quad D_{n+1} = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix}.$$

The number of (m, n) polygon mosaics is the $(0, 0)$ entry of A_n^m.

Now that we have enumerated polygon mosaics, we move to the more complicated polygon-free mosaics.

5 Counting Polygon-Free Mosaics

We use the following assignment for v to count polygon-free mosaics

$$\begin{cases} 7 & \text{for cells } \begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}, \begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix} \\ 0 & \text{for cells } \begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}, \begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix} \\ -1 & \text{for cells } \begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}, \begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix} \\ 1 & \text{otherwise.} \end{cases}$$

For example, if we let ℓ be the framed lattice on the right of Figure 3, we have $v(\ell) = -7^{13}$, as there are two $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ cells, one $\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}$ cells, twelve $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ cells, and one $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ cells.

We show our choice of v has useful properties by first considering the magnitude, then the sign. For the magnitude of v, for fixed m, n give an index $1 \leq i \leq N$ to every way one polygon can be formed in an (m, n) mosaic. A polygon is determined not only by its shape and orientation, but also its location within the mosaic. For example, Figure 4 depicts three distinct polygons in a $(2, 4)$ mosaic, which for reference we call polygons $1, 2$, and 3.

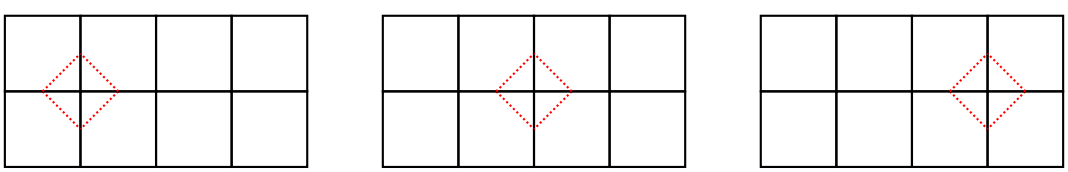


Figure 4: Polygons $1, 2$, and 3

Definition 5.1. Let M_i be the set of (m, n) mosaics that contain polygon i.

We immediately have the number of mosaics that contain at least one polygon is

$$7^{mn} - \left|\mathbb{P}^{(m,n)}\right| = \left|\bigcup_{i=1}^{N} M_i\right|. \tag{2}$$

Notice that not every subset of polygons has a mosaic that contains them all. For example, for the polygons in Figure 4 $M_1 \cap M_2 = M_2 \cap M_3 = \varnothing$ but $M_1 \cap M_3$ contains exactly the polygon mosaic with polygons 1 and 3.

Proposition 5.1. *For a polygon mosaic P, let $\ell = \mathcal{L}(P)$. If J is the subset of indices of the polygons in P we have*

$$\left|\bigcap_{j\in J} M_j\right| = \left|v(\ell)\right|. \tag{3}$$

Proof. From the definition of M_j, the left hand side of Equation 3 counts the number of mosaics that necessarily contain the polygons in J, and possibly contain other polygons, or tiles of any type that are not part of polygons. For the right hand side, if a cell comes from a tile that is part of a polygon, the absolute value of v of the cell is 1. Therefore, $|v(\ell)|$ is just 7 raised to the number of $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ cells in ℓ. As all tiles in $\mathbb{T}$ can map to $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$, $|v(\ell)|$ counts all possible tile assignments in all locations that are not part of a polygon in J. Therefore $|v(\ell)|$ counts mosaics that contain the polygons in J. This is the definition of $\bigcap_{i\in S} M_i$, and so the equality holds. □

For the sign of v, consider a polygon mosaic P, and let $\ell = \mathcal{L}(P)$. We show that although $v(\ell)$ is computed "cell-by-cell" of ℓ, we recover global information about P.

Proposition 5.2. *For a polygon mosaic P, let $\ell = \mathcal{L}(P)$. If P contains an even [resp. odd] number of polygons then $v(\ell)$ is positive [resp. negative].*

Proof. Let N be the number of vertices of ℓ that are 1, and define ℓ_0 to be the lattice formed by changing these N vertices from 1 to 0. That is, ℓ_0 has the same dimensions as ℓ but all vertices are 0. We will change ℓ_0 back to ℓ by flipping these vertices back from 0 to 1, one vertex at a time. The order in which we flip vertices is as follows.

We flip vertices one row at a time, working from top to bottom. In each row we make two passes, each time moving from left to right, and each time flipping a vertex if the corresponding vertex of ℓ is 1. On the first pass we flip a vertex if the vertex is directly above it is 1. (Because we are working the rows from top to bottom, this vertex is 1 in both our current lattice and also in ℓ.) On the second pass we flip the remaining vertices of the row, i.e., if the vertex is directly above it is 0.

This procedure creates a sequence $\ell_0, \ell_1, \ldots, \ell_N$ of lattices with $\ell_N = \ell$. Because $\ell = \mathcal{L}(P)$, we know ℓ is a proper framed lattice. Since every ℓ_i has the same dimensions as ℓ and has vertices of 0 in at least the same positions as in ℓ, we see every ℓ_i is a framed lattice. We will prove that every ℓ_i is also a proper lattice (which is the reason we make two passes through each row) and that by taking $P_i = \mathcal{L}^{-1}(\ell_i)$, the proposition holds for P_i and ℓ_i.

Because every ℓ_i is framed, in the transition from ℓ_{i-1} to ℓ_i, the single vertex that changes is adjacent to eight vertices, as in Figure 5, where the changing vertex # is 0 in ℓ_{i-1} and 1

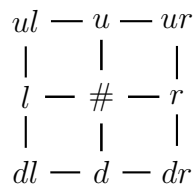


Figure 5: Cases Diagram

in ℓ_i, and the remaining vertices are labeled for their relative position to this central vertex, such as dl for down left and u for up.

We need to consider all possible adjacent vertices. Since our process goes from top to bottom one row at a time, we have $(dl, d, dr) = (0, 0, 0)$. Also, because of the way in which we make our two passes, if $u = 1$ then $r = 0$ and $(ul, l) \neq (0, 1)$, whereas if $u = 0$ then $(ur, r) \neq (0, 1)$. Additionally, suppose $u = 0$. Then neither (ul, l) nor (ur, r) equals $(1, 0)$. Indeed, if $(ul, l) = (1, 0)$ then since we are making the second pass through the row, we would have a cell $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ in ℓ_i and thus also in $\ell_N = \ell$, contradicting that ℓ is a proper lattice. Similarly, $(ur, r) = (1, 0)$ implies the that ℓ has a cell $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ (since with $ur = 1$, r cannot change in the second pass), again contradicting that ℓ is a proper lattice. Putting all this together, on the first pass, i.e., when $u = 1$, the vertices can only be as in the six cases pictured in Figure 6. Similarly, on the second pass we have $u = 0$ and one of the six cases of Figure 7.

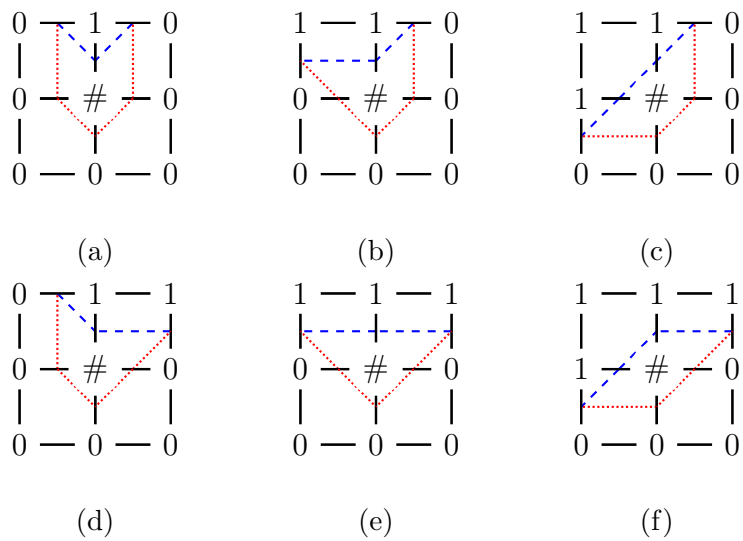


Figure 6: When $u = 1$

Since trivially ℓ_0 is a proper lattice, the twelve cases in Figures 6 and 7 show inductively that every ℓ_i is a proper lattice. For every i, we have already seen that ℓ_i is a framed lattice, so let $P_i = \mathcal{L}^{-1}(\ell_i)$. Figures 6 and 7 also show the portions of P_{i-1} and P_i that are in these two cells, where the dashed lines are from the polygons of P_{i-1} and the dotted lines from the polygons of P_i. (Recall that # is 0 in ℓ_{i-1} and 1 in ℓ_i.)

Let us say a cell is *negative* if it is $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$, i.e., if its evaluation under v is negative. The proposition can be rephrased by saying that the number of negative cells in ℓ and the

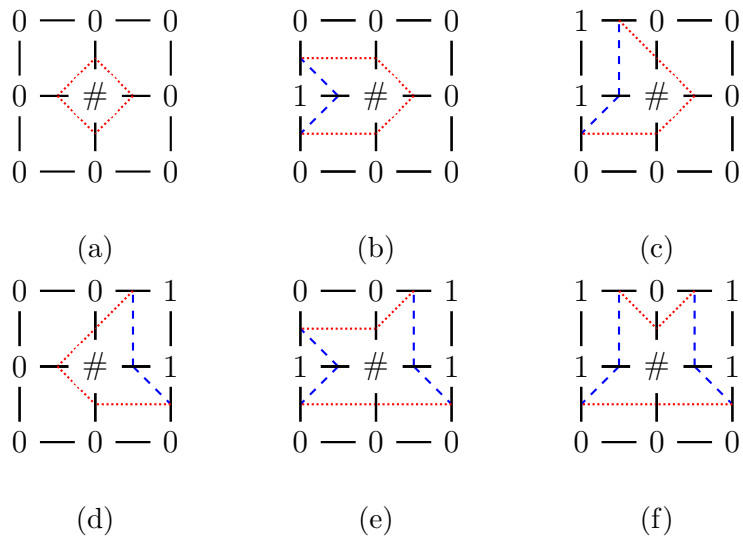


Figure 7: When $u = 0$

number of polygons in P have the same parity, i.e., are both even or both odd. For example, ℓ_0 satisfies the proposition because ℓ_0 has no negative cells and P_0 has no polygons.

Assume now that some P_{i-1} and ℓ_{i-1} satisfies the proposition, and let us show that the same is true for P_i and ℓ_i. This will complete the proposition. We know that we are in one of the twelve cases of Figures 6 and 7, and we must look at the number of negative cells in ℓ_i versus ℓ_{i-1}, and the number of polygons in P_i versus P_{i-1}. An easy case is (a) of Figure 7 because ℓ_i has one more negative cell (specifically, a cell $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$), and P_i has one more polygon than P_{i-1}.

Let us now look in the remaining cases at how the number of negative cells changes from ℓ_{i-1} to ℓ_i. In the six cases of Figure 6 and also case (d) of Figure 7, the four pictured cells have no negative cells regardless of whether # equals 0 or 1, so there is no change. The number is also unchanged in (b) of Figure 7 because in this case a single negative cell $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ changes position between ℓ_{i-1} and ℓ_i. In (c) of Figure 7, the number of negative cells increases by 2 because changing 0 to 1 in the central position creates both a cell $\begin{smallmatrix}1&0\\1&1\end{smallmatrix}$ and a cell $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$. For the final two cases, the number decreases by 1 in (e) of Figure 7 and increases by 1 in (f) of Figure 7. Therefore, we can complete the proposition by showing that in the six cases of Figure 6 and in (b), (c), (d) of Figure 7, P_{i-1} and P_i have the same number of polygons, whereas in (e) and (f) of Figure 7, the number of polygons in P_{i-1} and P_i differ by 1.

Consider, for example, (c) of Figure 6, and look at the point just above the lower left corner and the point just left of the upper right corner. In P_{i-1} these are connected by a straight dashed line, and in P_i these are connected by dotted line segments. We call both of these *inside connections*. Because P_{i-1} and P_i are polygon mosaics, these same points are connected by straight line segments outside the pictured area, and we will call these *outside connections* because they connect points on the edge of the pictured square with a path outside the square. This outside connection is the same in P_{i-1} and P_i. Therefore P_{i-1} and P_i have the same number of polygons because the polygon with this outside connection

changes shape, and all other polygons are unchanged. Similar reasoning completes the remaining cases of Figure 6 and cases (b), (c), (d) of Figure 7.

We now must show that in (e) and (f) of Figure 7, the number of polygons in P_{i-1} and P_i differ by 1. We will explicitly show this in (e), and essentially identical reasoning applies to (f). There are four places where inside connections meet the edge of the outer square. Beginning in the lower left and working clockwise, let x_1, x_2, x_3, and x_4 be these points. For example, x_3 is just left of the upper right corner and x_4 is just above the lower right corner. In P_{i-1}, there is an inside connection between x_1 and x_2, and another between x_3 and x_4, whereas P_i has inside connections between x_1 and x_4, and between x_2 and x_3.

Any outside connections in P_{i-1} and P_i must be identical, and since these are polygon mosaics, each of x_1, x_2, x_3, and x_4 is the end of some outside connection. One possibility is that we have one outside connection between x_1 and x_2, and another between x_3 and x_4. In this case, the inside and outside connections of P_{i-1} complete two polygons, one including x_1 and x_2, and the other including x_3 and x_4. However, P_i will have one fewer polygon because a single polygon contains all x_i. Another possibility is to have one outside connection between x_1 and x_4, and another between x_2 and x_3. In this case, in P_{i-1} a single polygon includes all x_i, whereas in P_i they complete two polygons, one containing x_1 and x_4, and the other containing x_2 and x_3, and therefore P_i has one more polygon than P_{i-1}. These are the only possibilities since it is impossible to have an outside connection between x_1 and x_3, and another between x_2 and x_4. This is because our polygons are in a plane, and polygons in a mosaic are not able to cross one another. This completes our proof. □

Theorem 5.3. *The number of (m,n) polygon-free mosaics is*

$$\left|\mathbb{P}^{(m,n)}\right| = \sum_{\ell} v(\ell), \tag{4}$$

where the sum is over all (m,n) framed lattices.

Proof. We begin by rewriting Equation 2 to

$$\left|\mathbb{P}^{(m,n)}\right| = 7^{mn} - \left|\bigcup_{i=1}^{N} M_i\right|.$$

By the inclusion-exclusion principle (see the Appendix), we have

$$-\left|\bigcup_{i=1}^{N} M_i\right| = \sum_{\varnothing \neq J \subseteq \{1,2,\ldots,N\}} (-1)^{|J|} \left|\bigcap_{j\in J} M_j\right|. \tag{5}$$

For Equation 5, for some $\varnothing \neq J \subseteq \{1,2,\ldots,N\}$, if $\bigcap_{j\in J} M_j$ is empty then the $\left|\bigcap_{j\in J} M_j\right|$ term is 0. Therefore, the domain of summation can be rewritten to be over all J such that there exists a polygon mosaic P with exactly the polygons in J. For fixed J, if $\mathcal{L}(P) = \ell$, Propositions 5.1 and 5.2 give the following

$$(-1)^{|J|}\left|\bigcap_{j\in J} M_j\right| = v(\ell).$$

Therefore, we can rewrite the sum in Equation 5 as

$$\sum_{J\subseteq\{1,2,\ldots,N\}} (-1)^{|J|}\Big|\bigcap_{j\in J} M_j\Big| = \sum_{\ell} v(\ell).$$

where ℓ is over all framed proper lattices that $\mathcal{L}^{-1}(\ell)$ has at least one polygon. Additionally notice that v applied to the binary lattice made up of all $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ cells equals 7^{mn}, and is the only framed proper lattice such that $\mathcal{L}^{-1}(\ell)$ has no polygons. This gives

$$\left|\mathbb{P}^{(m,n)}\right| = 7^{mn} - \Big|\bigcup_{i=1}^{N} M_i\Big| = \sum_{\ell} v(\ell),$$

where the sum is over all framed proper lattices. Finally, notice that the sum $\sum_\ell v(\ell) = 0$ for the framed lattices that contains $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$, as $v\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right) = v\left(\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right) = 0$. Therefore, summing over all framed lattices in Equation 4 gives the desired result. □

Theorem 5.3 let's us apply Theorem 4.2 to the v assignment at the beginning of this section, which gives the main result of the paper.

Theorem 5.4. *For positive integers n let A_n, B_n, C_n, D_n be $2^{n-1}\times 2^{n-1}$ matrices with integer entries, where $A_1 = \begin{pmatrix}7\end{pmatrix}, B_1 = \begin{pmatrix}-1\end{pmatrix}, C_1 = \begin{pmatrix}1\end{pmatrix}, D_1 = \begin{pmatrix}1\end{pmatrix}$ and*

$$A_{n+1} = \begin{pmatrix}7A_n & B_n\\ C_n & D_n\end{pmatrix} \quad B_{n+1} = \begin{pmatrix}-A_n & B_n\\ 0C_n & D_n\end{pmatrix}$$
$$C_{n+1} = \begin{pmatrix}A_n & 0B_n\\ C_n & D_n\end{pmatrix} \quad D_{n+1} = \begin{pmatrix}A_n & -B_n\\ C_n & 7D_n\end{pmatrix}.$$

The number of (m,n) polygon-free mosaics is the $(0,0)$ entry of A_n^m.

Finally, we introduce possible extensions to our work, as well as how our work relates to other mosaic enumeration efforts.

6 Related Work and Extensions

In terms of related work, our work can be seen as an extension to Hong and Oh [4] and further generalizing the techniques in Oh et al. [11] to explore a new direction in mosaic enumeration. To begin, Hong and Oh [4] provided bounds on the number of polygon mosaics.

Theorem 6.1 ([4])**.** *The number of (m,n) polygon mosaics for $m,n\geq 3$ is bounded between $2^{m+n-3}\left(\frac{17}{10}\right)^{(m-2)(n-2)}$ and $2^{m+n-3}\left(\frac{31}{16}\right)^{(m-2)(n-2)}$.*

Our Theorem 3.1 nicely compliments this result with a procedure to exactly enumerate polygon mosaics, as well as exactly enumerating the sequence A181245 on the OEIS [6]. It is important to note that contrary to this work, the authors in [4] did not consider the mosaic of all T_0 a polygon mosaic, and so our enumeration differs to theirs by 1. We also point out that our work allows for both polygon and polygon-free mosaics to be enumerated. The

methods in [11] could be used to enumerate just polygon mosaics (ie. an analagous result to Theorem 3.1), a fact which is likely known to the authors of [4] and [11], though to our knowledge has not appeared in print.

In related work, Lomonaco and Kauffman [5] introduced mosaics constructed from a tile set of 11 distinct tiles, of which $\mathbb{T}$ is a subset. The authors were interested in a subset of mosaics which they call *knot mosaics*. Oh et al. [11] enumerated the number of knot mosaics using an analagous method to that of Section 4.

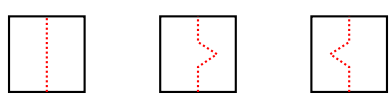
Theorem 6.2 ([11])**.** *The number of (m, n) knot mosaics for $m, n \geq 2$ is $2\,\|(X_{m-2} + O_{m-2})^{n-2}\|$, where $X_0 = O_0 = \begin{pmatrix}1\end{pmatrix}$ and X_{m-2} and O_{m-2} are $2^{m-2} \times 2^{m-2}$ matrices defined as*

$$X_{k+1} = \begin{pmatrix} X_k & O_k \\ O_k & X_k \end{pmatrix} \text{ and } O_{k+1} = \begin{pmatrix} O_k & X_k \\ X_k & 4O_k \end{pmatrix},$$

for $k = 0, 1, \ldots, m-3$. Here $\|N\|$ denotes the sum of elements of matrix N.

Oh and colleagues go beyond enumeration by bounding the growth rate of knot mosaics [8, 10, 1], and Oh further adapts techniques similar to Section 4 to solve problems in monomer and dimer tilings [7, 9]. Related ideas were independently used to enumerate the number of rectangular partitions of a rectangle in [12] for the sequence A182275 on the OEIS [6].

Enumerating the number of polygon-free mosaics in certain extensions of the tileset $\mathbb{T}$ is simple to do. For example, let's extend $\mathbb{T}$ to include three types of tiles that connect the top and bottom midpoints like below:



To enumerate polygon [resp. polygon-free] mosaics in this new tile set, one simply needs to adjust the assignments of v in Section 3 [resp. 5] to have $v\left(\begin{smallmatrix}0&1\\0&1\end{smallmatrix}\right) = v\left(\begin{smallmatrix}1&0\\1&0\end{smallmatrix}\right) = 3$.

We point out it is not possible to extend our results naively to the Lomonaco and Kauffman [5] tile set to enumerate the number of *knot-free mosaics*. This is specifically due to the deductions for Case 7e and Case 7f in the proof of Theorem 5.2, as knots are able to cross each other. This allows for additional subcases in which the number of knots does not change in these cases.

7 Acknowledgements

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8 Appendix

Proposition: The Inclusion-Exclusion Principle: For finite sets $A_1, A_2, \ldots, A_M$,

$$\left|\bigcup_{j=1}^{M} A_j\right| = \sum_{\emptyset \neq I \subseteq \{1,2,\ldots,M\}} (-1)^{|I|-1} \left|\bigcap_{j \in I} A_j\right|$$

Proof. The $M = 1$ case is trivial, and $M = 2$ case is the familiar and easily justified statement that $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$. Proceeding inductively, for $M \geq 3$, assume the result for $M - 1$ sets, and let us prove it for M sets. Let us split up the sum on the right into those sets I that do not contain M and those that do. By our inductive hypothesis, the first sum equals $|\bigcup_{j=1}^{M-1} A_j|$ and the second sum can be written as $\sum_{\{M\} \subseteq I \subseteq \{1,2,\ldots,M\}} (-1)^{|I|-1} \left|\bigcap_{j \in I} A_j\right|$. Separating out the term with $I = \{M\}$, this equals $|A_M| - \sum_{\emptyset \neq I \subseteq \{1,2,\ldots,M-1\}} (-1)^{|I|} |\bigcap_{j \in I} (A_j \cap A_M)|$. Again using our inductive hypothesis, the sum here is $\left|\bigcup_{j=1}^{M-1} (A_j \cap A_M)\right| = \left|\left(\bigcup_{j=1}^{M-1} A_j\right) \cap A_M\right|$. The entire right side of our desired equation therefore equals $\left|\bigcup_{j=1}^{M-1} A_j\right| + |A_M| - \left|\left(\bigcup_{j=1}^{M-1} A_j\right) \cap A_M\right|$. By the $M = 2$ case, this equals $|\bigcup_{j=1}^{M} A_j|$, as desired. $\square$